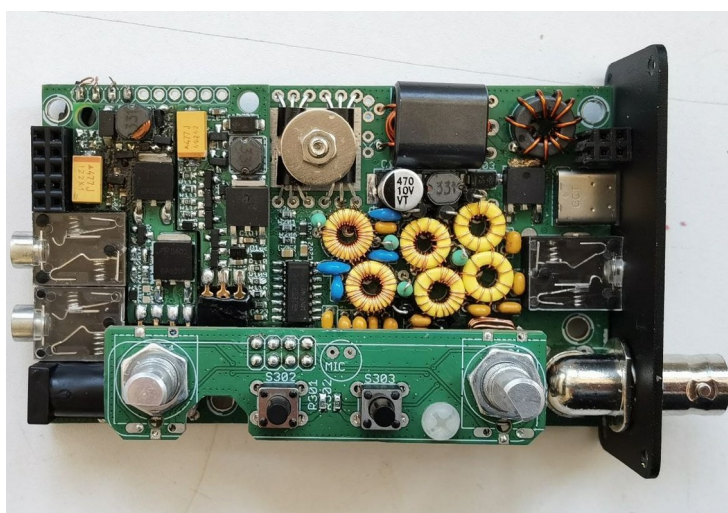


QMX: QRP Labs Multimode Xcvr (transceiver) Assembly manual

--- PCB Rev 4 / 5 ---



Contents

Contents.....	2
1. Introduction.....	3
2. Assembly.....	5
2.1 General guidelines.....	6
2.2 PCB diagrams (trace, component identification).....	11
2.3 Parts list.....	18
2.4 Inventory parts.....	25
2.5 Install all the ceramic capacitors.....	26
2.6 Install 1N4007 diodes.....	29
2.7 Install 47uH inductors.....	30
2.8 Install BS170 PA transistors.....	31
2.9 Assemble and install transformer T501.....	32
2.10 Prepare and install tapped inductor L401.....	35
2.11 Wind and install L502.....	44
2.12 Install Low Pass Filter toroids.....	45
2.13 Wind and install trifilar toroid T401.....	48
2.14 Wind and install transformer T507.....	51
2.15 Install connectors.....	52
2.16 Install pin header connector sockets (female).....	53
2.17 Install power supply boards.....	54
2.18 Install LCD module.....	58
2.19 Install 2x5-pin and 2x2-pin male pin header connectors.....	59
2.20 Install four 11mm nylon spacers.....	59
2.21 Install 20K trimmer potentiometer R305.....	60
2.22 Install 2x4-pin male header on controls PCB.....	61
2.23 Install rotary encoders.....	62
2.24 Install tactile switch buttons.....	63
2.25 Install electret microphone.....	63
2.26 Install 11mm nylon hex spacer.....	64
2.27 Fit Controls PCB to main PCB.....	65
2.28 Plug together the boards.....	65
2.29 QMX GPS interface and PTT output.....	67
2.30 FINAL CHECKS BEFORE APPLYING POWER THE FIRST TIME!.....	69
2.31 Connections for basic operation.....	70
2.32 Firmware installation.....	71
2.33 Optional enclosure.....	75
3. Resources.....	78
4. Document Revision History.....	78

1. Introduction

The QMX is a high performance, five-band 5W CW and Digital modes transceiver with CAT control and built-in USB sound card.

- Three versions: 80, 60, 40, 30, 20m, 60, 40, 30, 20, 17, 15m and 20, 17, 15, 12, 11, 10m
- 5W from a 9V or 12V supply (build options)
- Clean single signal output (zero residual carrier, zero unwanted sideband)
- Sunlight viewable, large 16 x 2 character yellow/green LCD, switchable backlight
- Built-in SWR metering and protection
- Lightweight: 210 grams
- Solid state PIN-diode switched Low Pass Filters and solid state Band Pass Filters
- Solid state transmit/receive switching
- High performance embedded-SDR SSB receiver using 112dB 24-bit stereo ADC chip
- Built-in USB sound card: 48ksps 24-bit stereo
- Built in USB Virtual COM port serial for CAT control
- Si5351A Synthesized local oscillator with better than 0.001Hz resolution and high precision 25MHz TCXO reference as standard
- Built-in signal generator
- Built-in suite of configuration and analysis tools
- Built-in iambic keyer
- On-screen S-meter
- CW decoder
- CW: Full or semi QSK, fast break-in; VFO A/B/Split, RIT, configurable offset, sidetone
- Beacon mode: standalone CW, FSKCW or WSPR operation
- GPS interface for real-time clock and frequency discipline for WSPR operation
- Lifetime free firmware upgrades with QRP Labs Firmware Update (QFU) bootloader for easy firmware update on any OS with no extra software, or drivers, or programming hardware
- All SMD pre-installed by factory, only through-hole component soldering by the constructor
- Low receive current (switching buck converters) 80mA (12V supply).
- Transmit current 1.0 – 1.1A (transmit dependent on power supply; example is approximately for 9V supply, 5W output; 12V supply TX current is proportionately lower).
- Connectors: USB-C (audio and serial for CAT), Power, PTT (external amp), audio out, paddle and RF
- Optional smart aluminium extruded enclosure measuring just 95 x 63 x 25mm

No test equipment is required to build, align and operate this digi modes transceiver. All adjustments can be made using built-in test equipment; or left alone (the transceiver will work well even with no adjustment).

We hope you enjoy building and operating this kit! Please read this manual carefully, and follow the instructions step by step in the recommended order. The construction is necessarily very compact and so it is important to really give attention to ALL the details described in this manual.

The circuit theory is described in detail in another document and we recommend reading and understanding this section too, to get the maximum enjoyment and education from your new radio.

The operating manual is a separate document and will get you started with QMX with CW, or with your WSJT-X or other digi modes software, in minutes.

Qty	Description	Component numbers
4	SMD: BSS84 MOSFET	Q102, Q202, Q204, Q502
12	SMD: BSS123 MOSFET	Q101, Q104, Q106, Q107, Q110, Q203, Q301, Q501, Q508, Q509, Q510, Q511
5	SMD: BC817 NPN SOT23-3	Q12, Q13, Q14, Q112, Q113
2	SMD: BC857 PNP SOT23-3	Q15, Q16
4	BS170: TO92 MOSFET	Q503, Q504, Q505, Q506
3	SMD: AO3415A SOT23-3	Q108, Q109, Q111
3	SMD: AOD403 (TO252)	Q103, Q105, Q507
1	SMD: 25MHz TCXO module	TCXO

Inductors

(see below for listing of band-version dependent inductors)

Qty	Description	Component numbers
5	SMD: 47uH L2012	L201, L202, L402, L503, L504
3	SMD: 330uH (CD54)	L101, L102, L501
4	47uH axial inductor	L505, L507, L509, L514
2	FT37-43 (dull black)	L502, T401
1	BN61-202 binocular, 3:3	T501
1	BN43-1502 binocular	T507

Miscellaneous

Qty	Value	Description
1	2.1 mm DC	2.1 x 5.5 barrel DC power connector
1	USB C	USB type C connector
1	BNC	Metal RF connector
3	3.5mm jack	3.5mm Stereo connector (PTT)
2	2x4	Pin header connector socket (female)
1	2x4	Pin header connector 90-degree plug (male)
1	2x3	Pin header connector socket (female)
1	2x3	Pin header connector 90-degree plug (male)
1	2x5	Pin header connector socket (female)
1	2x2	Pin header connector socket (female)
1	2x4	Pin header connector 17mm long plug (male)
1	2x2	Pin header connector plug (male)
1	2x5	Pin header connector plug (male)
1	PCB	Main PCB, 92 x 139.4 mm
1	1602 LCD	80x36mm HD44780 LCD module yellow/green
1	6mm Electret	Microphone
2	Rotary	Rotary encoder switch

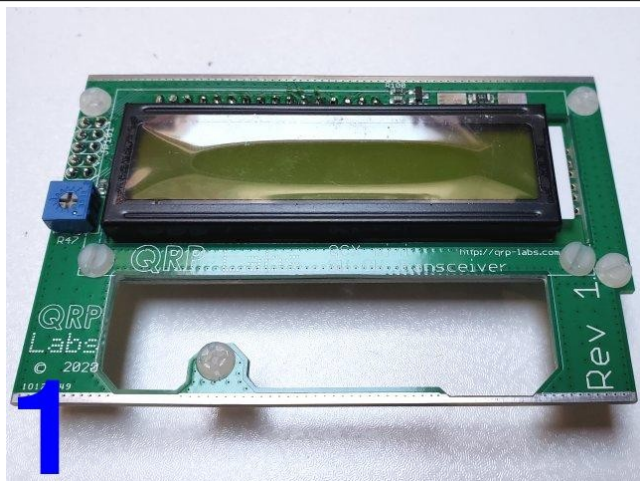
Qty	Value	Description
2	6x6x8mm	Push-button switch
2	15mm	Knob
1	365 cm (80-20m version)	0.33 mm diameter wire (AWG #28) (varies depending on QMX version as noted)
1	65 cm (80-20m version)	0.60 mm diameter wire (AWG #22) (varies depending on QMX version as noted) NOTE: Recent kits are supplied with only 0.33mm wire for everything, it was found to be easier to handle and slightly increase performance!
1	M3 10 mm	Steel 10mm long M3 screw
1	M3	Steel M3 nut
1	M3 12 mm	Steel 12mm diameter M3 washer
5	M3 x 11mm	Nylon threaded hex spacer
12	M3 x 6mm	Nylon screw
2	M3 x 9mm	Nylon screw
6	M3	Nylon nut

Enclosure (OPTIONAL)

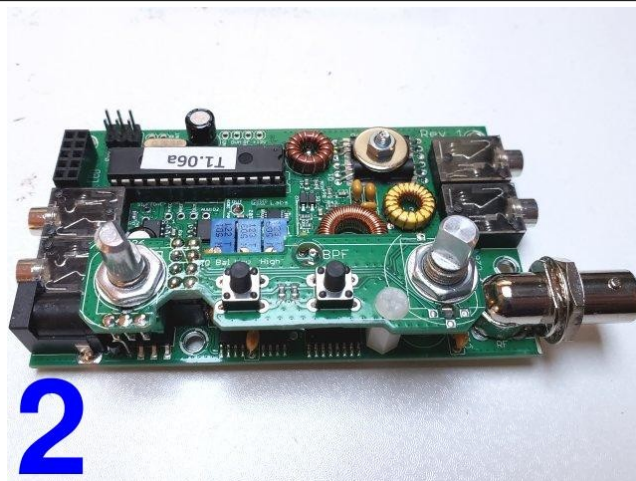
Qty	Value	Description
1	Top	Extruded aluminium top, cut, drilled and laser-etched
1	Bottom	Extruded aluminium bottom
1	Left panel	Laser etched, drilled for three connectors
1	Right panel	Laser etched, drilled for three connectors
8	M2.5 machine screw	Screws to secure end panels
4	Rubber foot	Self-adhesive rubber foot

2.33 Optional enclosure

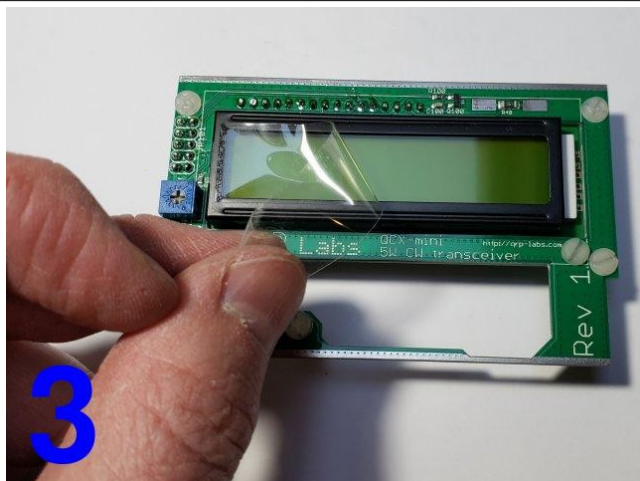
Installation in the QMX enclosure is simple and requires no wiring. It is important to do the assembly in the correct sequence, as follows (Photographs are from the similar QCX-mini).



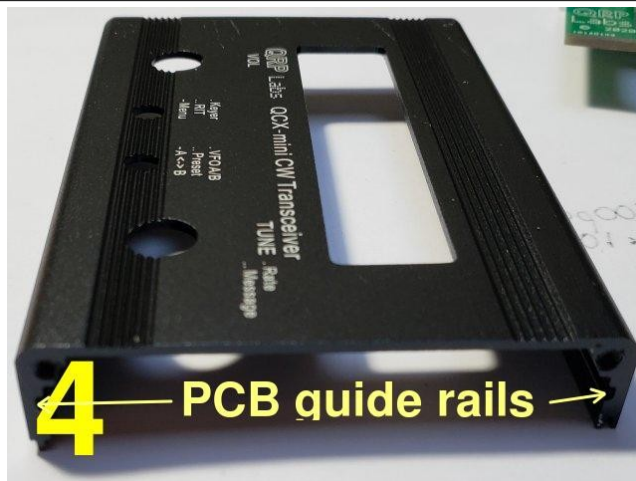
Start with the Display board...



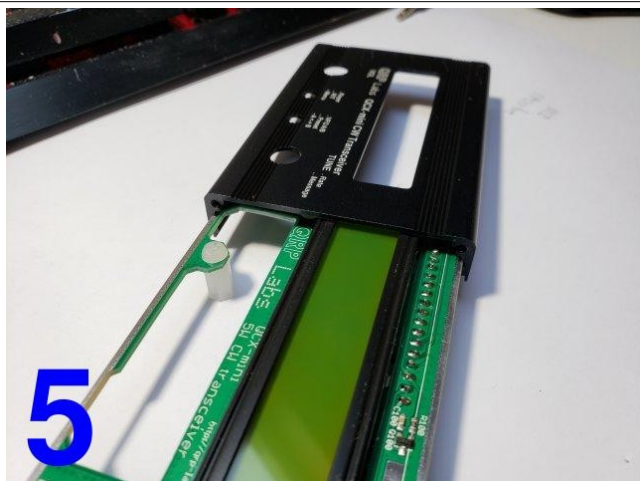
... and the main board, with the controls board bolted in position.



Peel off the plastic protective coating from the LCD module.



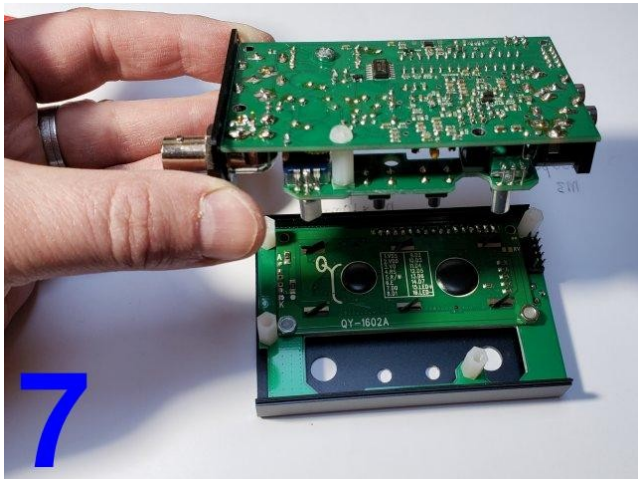
Here's the top half of the enclosure. Note the PCB guide rails in the extrusion walls.



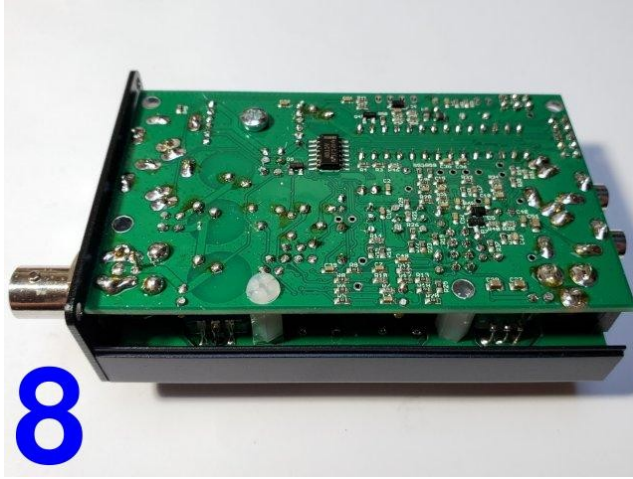
Slide the display board into position along the PCB guide rails in the enclosure.



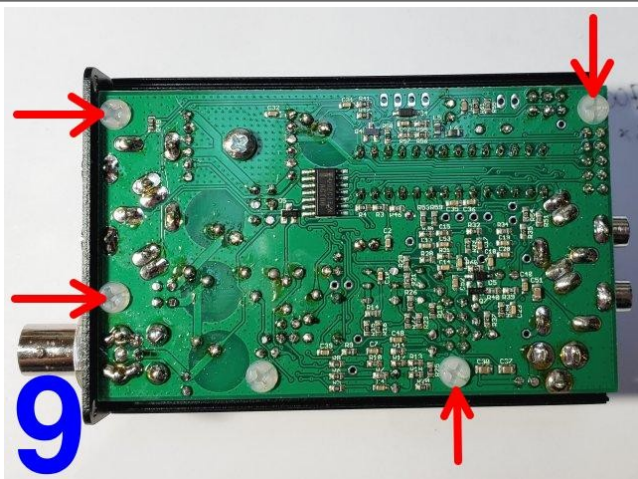
Bolt the right-hand side panel to the main PCB using the supplied BNC washer and nut



7 Place the front of the enclosure face down on the bench as shown, and prepare to attach the main board.



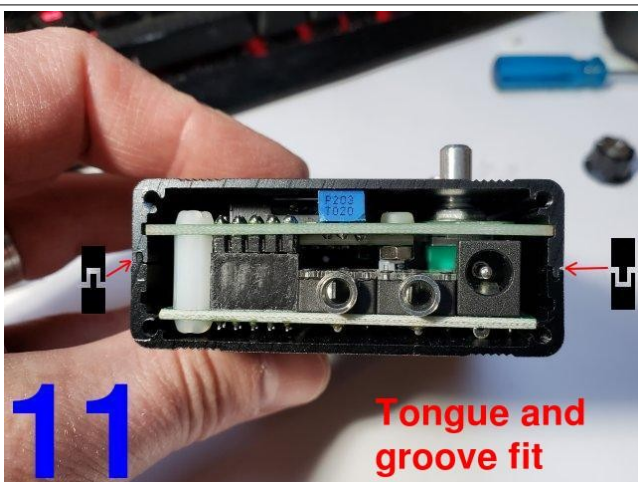
8 Align the 2x5-pin header connector between the main and display boards; some wiggling will be needed to get the controls to fit through the holes in the front panel.



9 Fit four M3 6mm nylon screws in the positions shown



10 This is how it looks from the DC connector end



11 Determine correct orientation of the bottom half of the enclosure; note the tongue-and-groove arrangement which means that the bottom half only fits one way round! Make sure you have the correct way.



12 Now bolt the left-hand side panel to the enclosure extruded top and bottom halves using four of the supplied small black countersunk screws in the panel corners. The screws need to be carefully aligned and should screw in easily (if properly aligned).



Screw in the other four black screws in the corners of the right-hand side panel.



Apply the supplied four self-adhesive feet to suitable positions on the base of the enclosure if desired (optional).



Install the supplied knobs by tightening their grub screws. Leave a small gap between each knob and the front panel, to allow the knob to turn easily and be pressed down to activate the shaft button. The grub screw of the knobs is best sited on the ROUND part of the shaft not the flat path; the knob will be less prone to wobble or be misaligned.



Plug in the power...

SUCCESS!

(hopefully)

Disassembly of the QMX should follow similar steps, in reverse.

3. Resources

- For updates and tips relating to this kit please visit the QRP Labs QMX kit page <http://qrp-labs.com/qmx>
- For any questions regarding the assembly and operation of this kit please join the QRP Labs group, see <http://qrp-labs.com/group> for details

4. Document Revision History

1.00	04-May-2023	First draft
1.00a	24-May-2023	First official version
1.00b	25-May-2023	Some spelling mistake corrections and update to the page 59 photo
1.00c	25-May-2023	Clarification of diode polarity in section 2.6
1.00d	26-May-2023	Section 2.13 amendment (wire lengths) and page 20 BOM correction
1.00e	31-May-2023	Added first time checks section and other updates
1.00f	09-Jul-2023	Amended for modified L401 style (to remove 20m sensitivity problem); corrected error with the 180p and 270p capacitors in the LPF being swapped.
1.00g	18-Aug-2023	Added note about removing superfluous PCB rail material from manufacturing process. T1 wrongly named on page 4, changed to T501. Added more checks including Q103/Q104 short, before power up
1.00h	20-Aug-2023	Removed 22pF capacitor from BOM on p21 and p22 photo
1.00i	12-Sep-2023	P10 step 14, referred to 2.32, it is now correctly 2.33. P43, 10mm nylon nut should be (and now is) 9mm. Moved T501 preparation to a separate document. Removed the recommendation to drill out the SMPS spacer nuts Contrast adjustment pot corrected to R305
1.00j	24-Sep-2023	Corrected photograph caption in section 2.17 (no “de-threaded”)
1.01	28-Nov-2023	Manual for Rev 2 PCB QMX assembly
1.02		Fix mix-up of red/yellow L401 for high/low band versions
1.02a	10-Feb-2024	Add note on page 45 regarding the grounded PTT modifications
1.04	08-May-2024	Manual for Rev 4 PCB QMX assembly (incl 60-15m version)
1.04a	03-Jun-2024	Corrected problem with 60-15m section 2.5 diagram, page 26
1.04b	29-Jul-2024	Recommended to fit grub screw at round part of shafts not flat
1.04c	24-Sep-2024	IC101 was incorrectly identified as 78M05, should be 78M33 Page 41, 20-10m section heading typo Fixed typo, L401 was incorrectly called L405 on 60-15m version